

INSTRUCTIONS: Circle the best answer (A, B, C, or D) for each question.

=====

SECTION A: FRACTIONS (Questions 1-6)

=====

1. Simplify: $12/18$
A) $1/3$
B) $2/3$
C) $3/4$
D) $4/6$
2. Which fraction is equivalent to 0.75?
A) $1/4$
B) $2/5$
C) $3/4$
D) $4/5$
3. Add: $1/4 + 2/3$
A) $3/7$
B) $3/12$
C) $11/12$
D) $5/6$
4. Multiply: $2/5 \times 3/4$
A) $5/9$
B) $6/20$
C) $3/10$
D) $1/2$
5. Divide: $3/4 \div 1/2$
A) $3/8$
B) $1/2$
C) $3/2$
D) $2/3$
6. A recipe uses $2/3$ cup sugar. You make half the recipe. How much sugar?
A) $1/6$ cup
B) $1/3$ cup
C) $1/2$ cup
D) $2/3$ cup

=====

SECTION B: ALGEBRA (Questions 7-12)

=====

7. Solve for x: $3x + 7 = 22$
A) $x = 3$
B) $x = 5$
C) $x = 7$
D) $x = 15$
8. Which is a quadratic equation?

- A) $2x + 1 = 9$
- B) $x^2 - 5 = 0$
- C) $1/x = 4$
- D) $x^3 + 2 = 0$

9. Solve: $x^2 - 9 = 0$

- A) $x = 3$ only
- B) $x = -3$ only
- C) $x = 3$ or $x = -3$
- D) $x = 9$ or $x = -9$

10. Factor: $x^2 + 7x + 12$

- A) $(x + 2)(x + 6)$
- B) $(x + 3)(x + 4)$
- C) $(x + 1)(x + 12)$
- D) $(x - 3)(x - 4)$

11. Solve: $2(x - 4) = 10$

- A) $x = 5$
- B) $x = 7$
- C) $x = 9$
- D) $x = 11$

12. What is the discriminant of $x^2 + 6x + 9 = 0$?

- A) -36
- B) 0
- C) 9
- D) 36

=====

SECTION C: GEOMETRY (Questions 13-18)

=====

13. The sum of angles in a triangle is:

- A) 90 degrees
- B) 180 degrees
- C) 270 degrees
- D) 360 degrees

14. A rectangle has length 12 cm and width 5 cm. Its area is:

- A) 17 cm^2
- B) 34 cm^2
- C) 60 cm^2
- D) 120 cm^2

15. The circumference of a circle with radius 7 cm is (use $\pi = 22/7$):

- A) 22 cm
- B) 44 cm
- C) 49 cm
- D) 154 cm

16. Two angles are supplementary. One is 65 degrees. The other is:

- A) 25 degrees
- B) 65 degrees
- C) 115 degrees
- D) 125 degrees

17. A triangle has base 10 cm and height 6 cm. Area =

- A) 16 cm^2
- B) 30 cm^2
- C) 60 cm^2
- D) 120 cm^2

18. A square has perimeter 36 cm. Side length =

- A) 6 cm
- B) 9 cm
- C) 12 cm
- D) 18 cm

=====

ANSWER KEY

=====

FRACTIONS

- 1. B $12/18 = 2/3$
- 2. C $0.75 = 3/4$
- 3. C $1/4 + 2/3 = 3/12 + 8/12 = 11/12$
- 4. C $2/5 \times 3/4 = 6/20 = 3/10$
- 5. C $3/4 \div 1/2 = 3/4 \times 2/1 = 3/2$
- 6. B Half of $2/3 = 1/3$ cup

ALGEBRA

- 7. B $3x = 15, x = 5$
- 8. B $x^2 - 5 = 0$ is degree 2
- 9. C $x^2 = 9, x = 3$ or -3
- 10. B $x^2 + 7x + 12 = (x + 3)(x + 4)$
- 11. C $2(x - 4) = 10, x - 4 = 5, x = 9$
- 12. B $D = 36 - 36 = 0$

GEOMETRY

- 13. B Triangle angle sum = 180 degrees
- 14. C Area = $12 \times 5 = 60 \text{ cm}^2$
- 15. B $C = 2 \pi r = 2 \times 22/7 \times 7 = 44 \text{ cm}$
- 16. C $180 - 65 = 115$ degrees
- 17. B Area = $1/2 \times 10 \times 6 = 30 \text{ cm}^2$
- 18. B $36 \div 4 = 9 \text{ cm}$

END OF DIAGNOSTIC